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Chapter 5

Enabling Existing Elemental Activities



Why Read This?



Tip: Read this if you have System 21 Activities which you need as part of a process and which are not currently Work Management enabled. Print Purchase Orders is used as a working example. Refer to the appendix called Enabled Activities List.

Enabling Activities Stop Press



Warning: Please take a minute to read this.

Before you start to enable any Elemental Activities please refer to the document on the installation CD called stopprss.doc. This document contains the very latest information to help you to enable existing Elemental Activities for Work Management.

This will help you to select the resources you need both for programming staff and hardware and software equipment and will give you helpful reports from people with experience of enabling.

The Two Stage Process

There are two stages involved in enabling existing System 21 activities for Work Management. Firstly you change the RPG program and secondly you edit the Properties in Constructor. However, by completing the Properties you are not making the activity form part of a Business Process. Activities can be, and often are, fully enabled for Work Management and yet form no part of a Business Process.



Tip: Remember: You enable Activities but you activate a Business Process containing activities. An activity can be enabled without being part of a Business Process.

There is a tutorial for each stage, both of them using the Print Purchase Order program (PM084) as an example.

Stage 1 Edit The RPG Program

At the beginning of the program you have to place a call to one of the subroutines in the Copy Source. This does three things:

- It tells the Work Management Engine that the activity is running
- It generates time stamps for the transaction
- It may pass back to the application the document to process.

At the end of the processing of each document, you have to place a call to one of the subroutines in the Copy Source to tell the Work Management Engine that, for this document, activity processing is complete and may progress on to the next stage.



Note: Standard subroutines are provided for you in an RPG Copy Source. Please refer to the appendix called Copy Source Code.



Note: All modifications to the program are marked with WF_001.

Tutorial For Stage 1

This tutorial explains how to enable the Print Purchase Order program (PM084).

This is what you do:

Step 1

Incorporate the standard copy source routine @WMNABLE@C in OSLCOPYSRC into the program as follows :

```
1575.00      * Copy Source for Work Management Enabling
1576.00      C/COPY OSLCOPYSRC/QRPGSRC,@WMNABLE@C
```

This provides all the sub-routines required for enabling Work Management in the program.

Step 2

Do the EWM active check using subroutine XAWMCH.

This must be done after the LDA has been defined and retrieved by the program so that L#JBNO is set before calling subroutine:

```
0359.00      * Check to see if Work Management is Enabled and Active
0360.00      C                                EXSR XAWMCH
```

Return value

If WFAFWM = '1' Work Management is active and the following steps will be executed.

Step 3

Identify whether there can be more than one Business Object involved in the activity. In this program this is possible as there may be several purchase orders currently waiting to print.

If order number is not passed in, the program tries to process all orders using the following lines. The subroutine GETORD is used to get the next order to be processed:

```
0406.00 B01  C          ORDN      IFEQ *BLANKS
0407.00      C                      EXSR GETORD
0408.00      C                      MOVE EOFORD      *INLR
0409.00      *
0410.00      C                      MOVE '1'          HEADER
0411.00      *
0412.00 E01  C                      END
```



Warning: Where a particular order is not specified in the parameter ORDN, you must retrieve Work Management information before locking each Business Object.

Use the subroutine XAWMBO as follows:

```
1516.00      * If Work Management is Enabled Get Work Mgmt Info for BPO
1517.00 B04  C          WFAFWM      IFEQ '1'
1518.00      C                      MOVE *BLANKS      P#IOBJ
1519.00      C                      MOVE 'PURORDER' P#IOBJ
1520.00      C                      MOVE *BLANKS      P#IOBR
1521.00      C                      MOVE LORDN02      P#IOBR
1522.00      C                      MOVE '1'          P#KTYP
1523.00      C                      MOVE *BLANK      WF52FS
```

```
1524.00      C                      EXSR XAWMBO
1525.00 E04  C                      ENDIF
```



Note: In the case where a particular order is specified in the parameter ORDN, only one Business Object will be processed and Work Management information will already have been retrieved by XAWMCH so a call to XAWMBO isn't required.

Step 4

Attempt to Validate and Lock the current Business Object and check result as follows :

```
0452.00      * If Work Mgmt is Enabled Validate & Lock Bus Process
0453.00 B01  C          WFAFWM      IFEQ '1'
0454.00      C                      EXSR XAWMLK
0455.00      * Check validation and locking OK
0456.00 B02  C          P#RTCD      IFNE '01'
0457.00      C                      GOTO MAIN10
0458.00 E02  C                      ENDIF
0459.00 E01  C                      ENDIF
```

Step 5

When the order is printed, complete the activity for the current Business Object. The section in this program where order printing is completed is the section identified by the MAIN50 tag. This is where the completion should be done as follows:

```
1075.00      * If Work Management is Enabled complete for the obj ref
1076.00 B01  C          WFAFWM      IFEQ '1'
1077.00      C                      EXSR XAWMCM
1078.00 E01  C                      ENDIF
```



Note: The two other subroutines in the Copy source were not required in this program (i.e. XAWMBK and XAWMWK).

XAWMBK would be used if backing out a transaction was required, for example in an interactive program where F3 or F12 were pressed. As this program is a program run in batch no call is necessary.

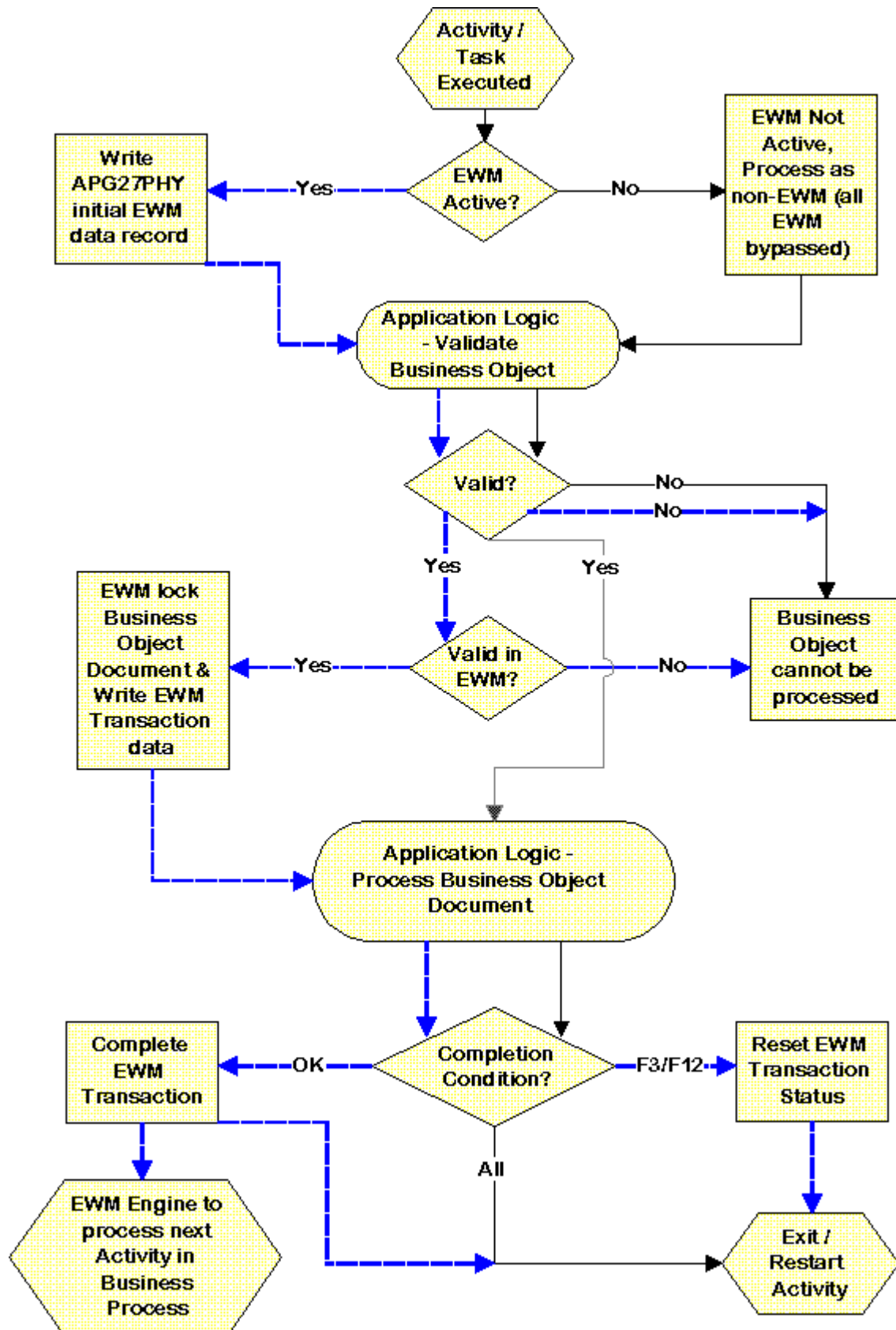
XAWMWK would be used if it was required to store completion information during processing. This isn't required in this program as I've used XAWMCM after each order is processed. I could have accumulated details using XAWMWK and once complete, called the subroutine again to trigger and complete updates (thereby getting rid of the need for XAWMCM call).



Warning: Check that the Activity Code defined in the RPG program source code is unique and matches the one on the Work Mgt panel on the Properties Window in Constructor.

Flowchart

The following diagram shows the application changes required in order to enable a task for use with JBA's Work Management solution. This is achieved by incorporating the standard copy source routine WMNABLE in the relevant programs, and using it appropriately. The diagram shows the main processes required for an application to link in to the EWM Engine.



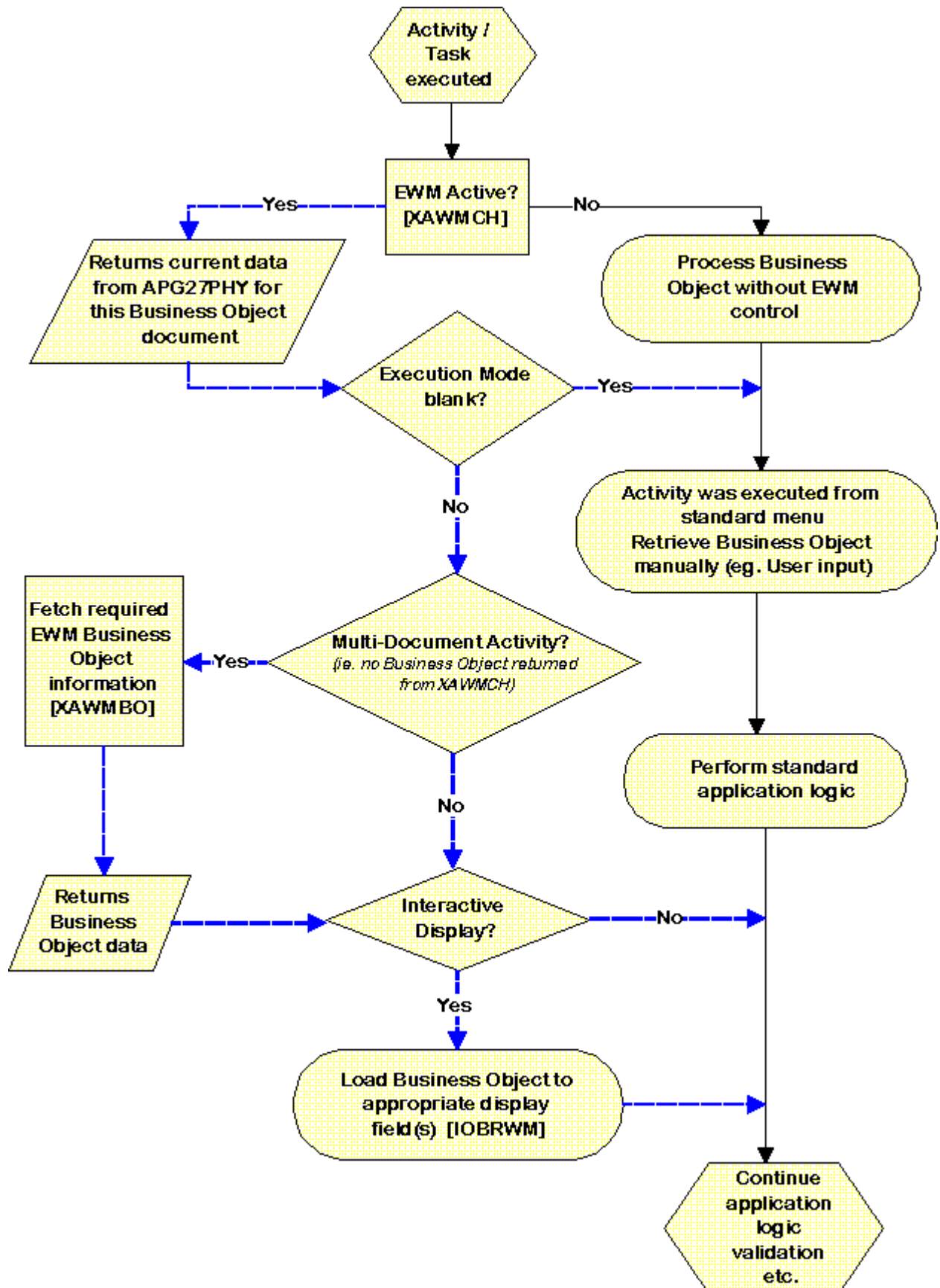
Initial Processing

Each application program wishing to use EWM, must as a minimum call the XAWMCH subroutine in the copy source. This identifies whether or not EWM should be used in this session or not. By minimising the EWM check to this one call, the overhead on Customers not using EWM at all should be acceptable. If a suite of application programs can use an internal control to minimise this further, so much the better.

XAWMCH will return the main detail required for a single business object (eg. pick note or order). The aim of returning the Business Object information is to populate existing application entry fields (eg. the Business Object reference Order '0001234' is output in the first screen of Order Acknowledgement Print if the activity was part of a Business Process for that Business Object).

However, if this is an initial activity for a business object (eg. order entry), then the information returned will be blank. In this case, normal application processing continues.

If there is more than one Business Object involved in this activity, then XAWMCH will return none. It is necessary for the application program to access XAWMBO for each Business Object document once they are known to the application, that is retrieved by program or entered by the User.



Task Initiation

When tasks are initiated through the standard application manager menu driver (XA541CLP), or via the Work Management driver (XA060CLP) application manager will write a record to the AM Workflow Details file (APG27PHY). This will only happen if tasks are executed in this manner. If jobs are submitted in any other way (e.g. XA902) then AM workflow details will not exist. This file is keyed by OSL job number and is used by all application programs in order to determine if Work Management is active, and if so, how processing should continue.

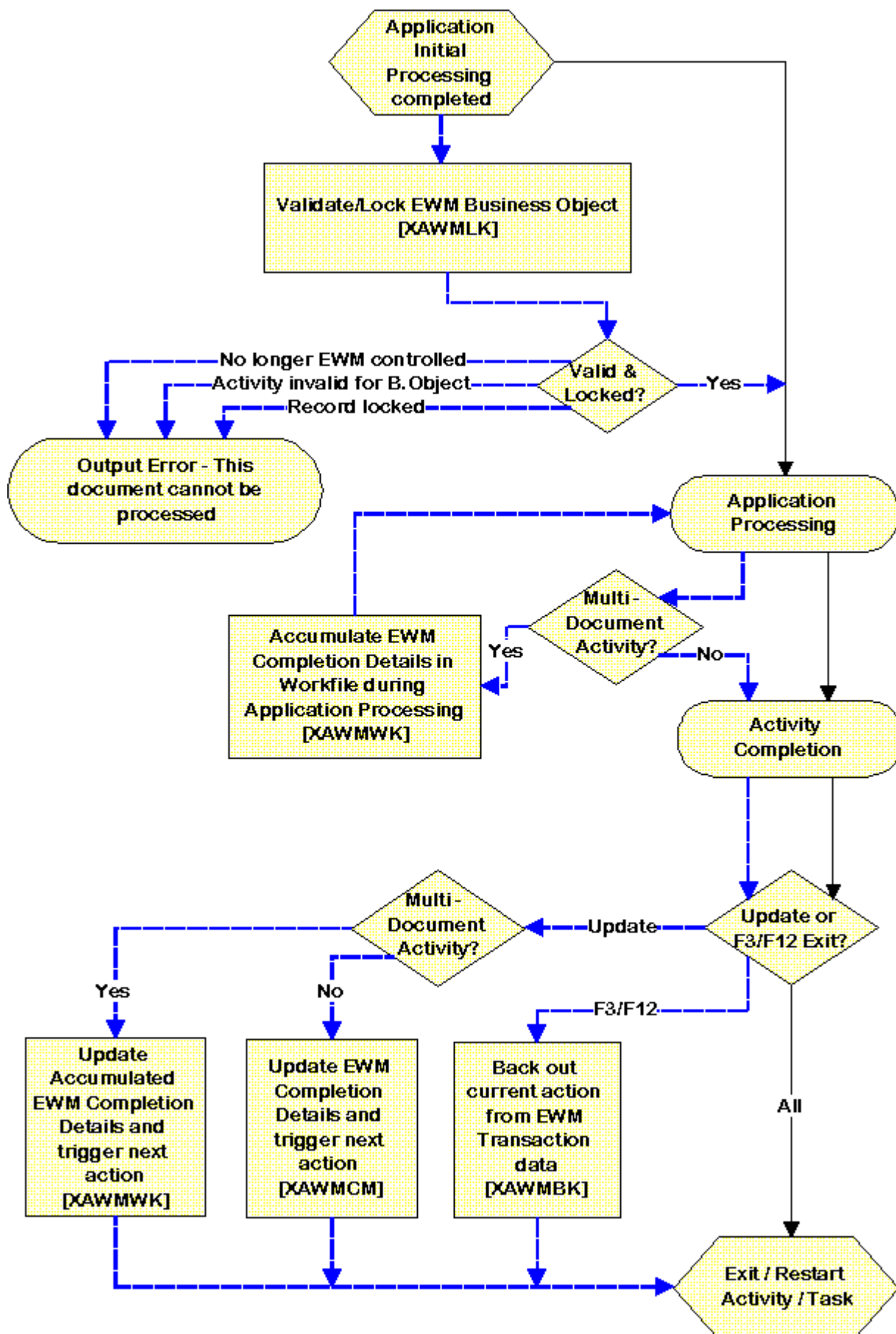
Validation, Locking And Completion

EWM Validation is handled by one subroutine call - XAWMLK. If valid, it automatically locks the Business Object within the EWM Engine. If invalid, it returns a code to the application program. It can be invalid due to an existing record lock, or due to no EWM transaction records existing (when this is not an initial activity). The latter suggests that at some stage, the transaction dropped out of EWM control (or was never in it). This would happen if introduced to the system via a non-EWM enabled activity. If the Business Object is under EWM control, it can only be processed by the correct activities within the controlling Business Process. If the Business Object is entered in an EWM enabled activity that it should not be in, or should not be in at that particular stage of the Business Process, the EWM Engine will detect it, and fail the validation.

Once passed validation, normal Application processing takes place. At the end of this, the EWM Engine needs to complete or back out the current action. To back out a transaction (eg. when F3 or F12 are taken), subroutine XAWMBK is used. To complete a transaction, XAWMCM is used.

If multiple documents are processed within the activity, it may be easier to store the EWM completion information during application processing (eg. when updating a subfile). Subroutine XAWMWK allows this. Once the accumulations are complete, and the updates are occurring, this subroutine should be called again to trigger and complete the updates. In this case, a call to XAWMCM is NOT required.

All three of these subroutines will cause any locks to be released, and the Update ones will trigger the EWM Engine to automate the next activity in the relevant Business Process for that document.



Stage 2 Edit The Properties In Constructor



Note: This could mean defining a new Elemental Activity or editing an existing activity definition. This tutorial shows you how to define a new Elemental Activity.

You use Constructor to add additional information about how the activity operates, such as Execution Modes, Output Nodes and Documents.

Tutorial For Stage 2

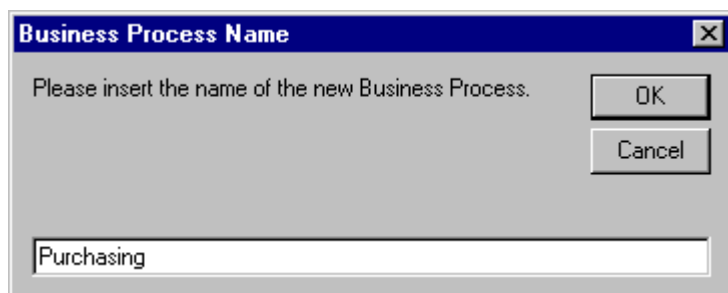
This tutorial explains how to enable the Print Purchase Order program (PM084). In Constructor this is called an Elemental Activity.



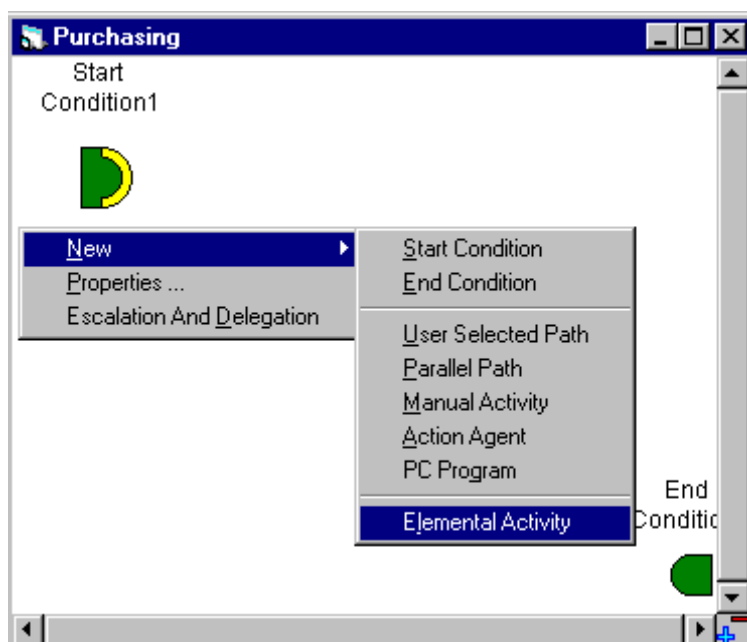
Note: This tutorial explains how to set up an Elemental Activity from beginning to end, including setting up a new Business Process in which to enclose the Elemental Activity. However, in reality it is likely that you might want to take an activity from the Component Browser Area, which is already known to Constructor, and edit its properties in order to enable it for Work Management.

This is what you do:

1. Launch Constructor.
2. Select from the File Menu, New\Business Process.



3. Give the Business Process a name. For example call it Purchasing. Then click OK. The Start Condition and End Condition are placed on the Workspace automatically.
4. Right click anywhere on the Workspace and select New/Elemental Activity from the pop up menu.



This displays the Properties Window with the following panels:

- General
- Details
- Advanced
- Work Mgt.



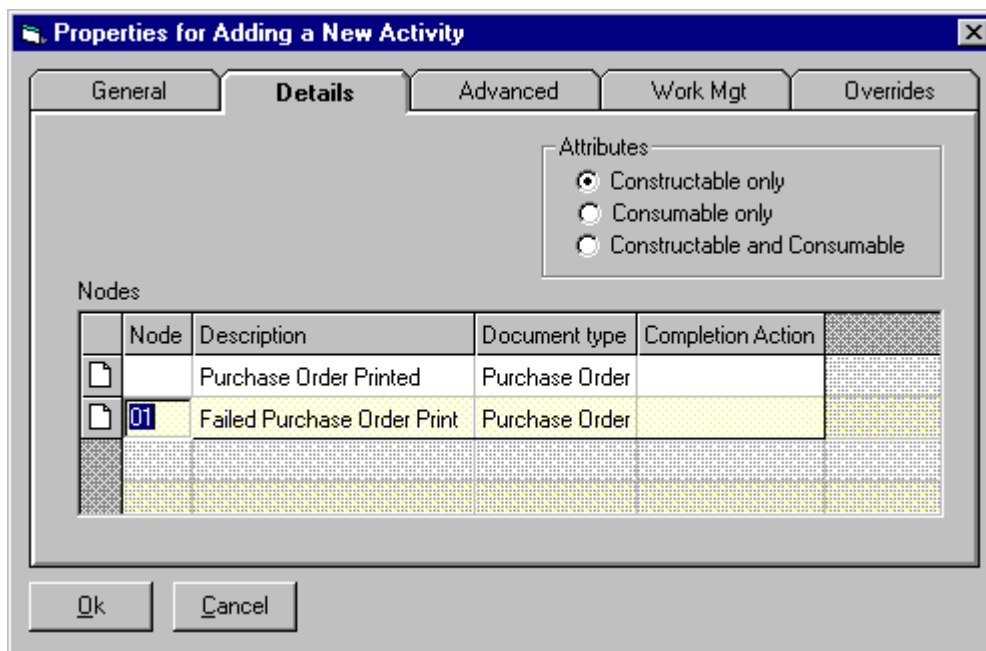
Note: If you are enabling an existing Elemental Activity, already known to Constructor and listed in the Component Browser Area, then you would see a fifth panel called Overrides. This is for overriding the name of an activity, only when it becomes listed in the Action List, for the Explorer user.

5. Complete the fields as follows:



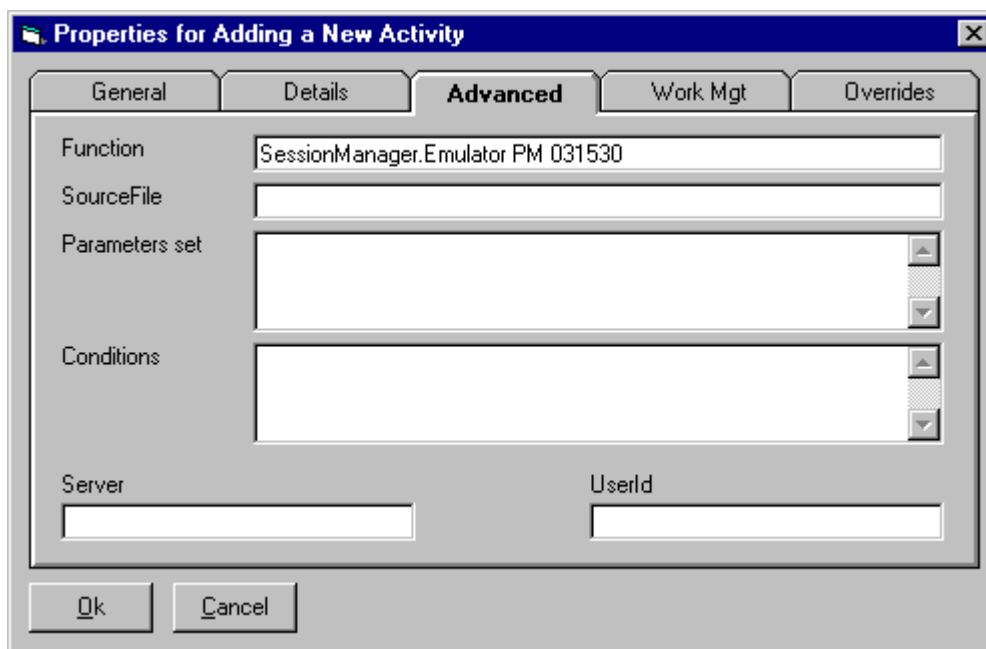
Warning: When you edit *Properties* you are updating the Design Time database for everyone.

The screenshot shows a dialog box titled "Properties for Updating Activity". It has five tabs: "General", "Details", "Advanced", "Work Mgt", and "Overrides". The "General" tab is active. Inside the "General" tab, there are three main sections: "Name" with a text box containing "Print Purchase Order", "Description" with a text box containing "Print Purchase Order" and a vertical scrollbar, and "Input Document Type" with a dropdown menu showing "Purchase Order". To the right of the dropdown is a checkbox labeled "Work Management Enabled" which is checked. At the bottom of the dialog are two buttons: "Ok" and "Cancel".



The 'Details' tab of the 'Properties for Adding a New Activity' dialog. It features a 'Nodes' table with two entries: 'Purchase Order Printed' and 'Failed Purchase Order Print'. The second entry is selected. Above the table, there are radio buttons for 'Attributes': 'Constructable only' (selected), 'Consumable only', and 'Constructable and Consumable'. 'Ok' and 'Cancel' buttons are at the bottom.

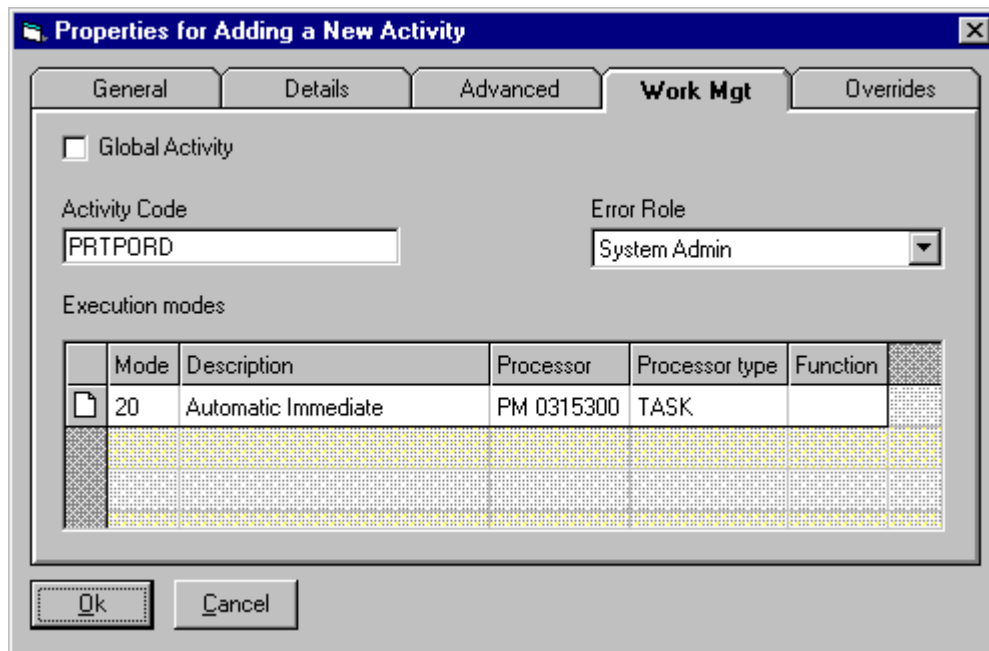
Node	Description	Document type	Completion Action
	Purchase Order Printed	Purchase Order	
	Failed Purchase Order Print	Purchase Order	



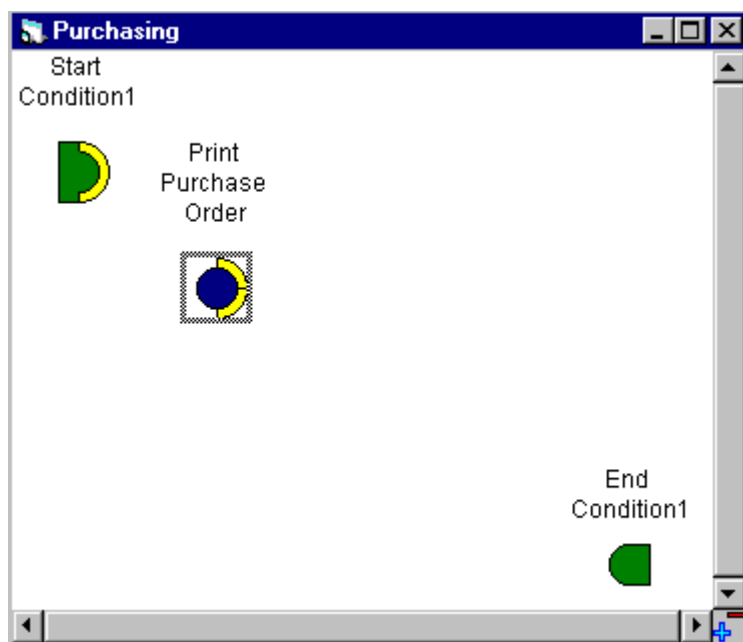
The 'Advanced' tab of the 'Properties for Adding a New Activity' dialog. It contains input fields for 'Function' (SessionManager.Emulator PM 031530), 'SourceFile', 'Parameters set', 'Conditions', 'Server', and 'UserId'. 'Ok' and 'Cancel' buttons are at the bottom.



Warning: The Activity Code on the Work Mgt panel must be unique and must match the one defined in the RPG program source code.



6. You can ignore the Overrides Panel. Click OK to return to the Workspace.



7. Select File and Save.

This is the end of the tutorial.